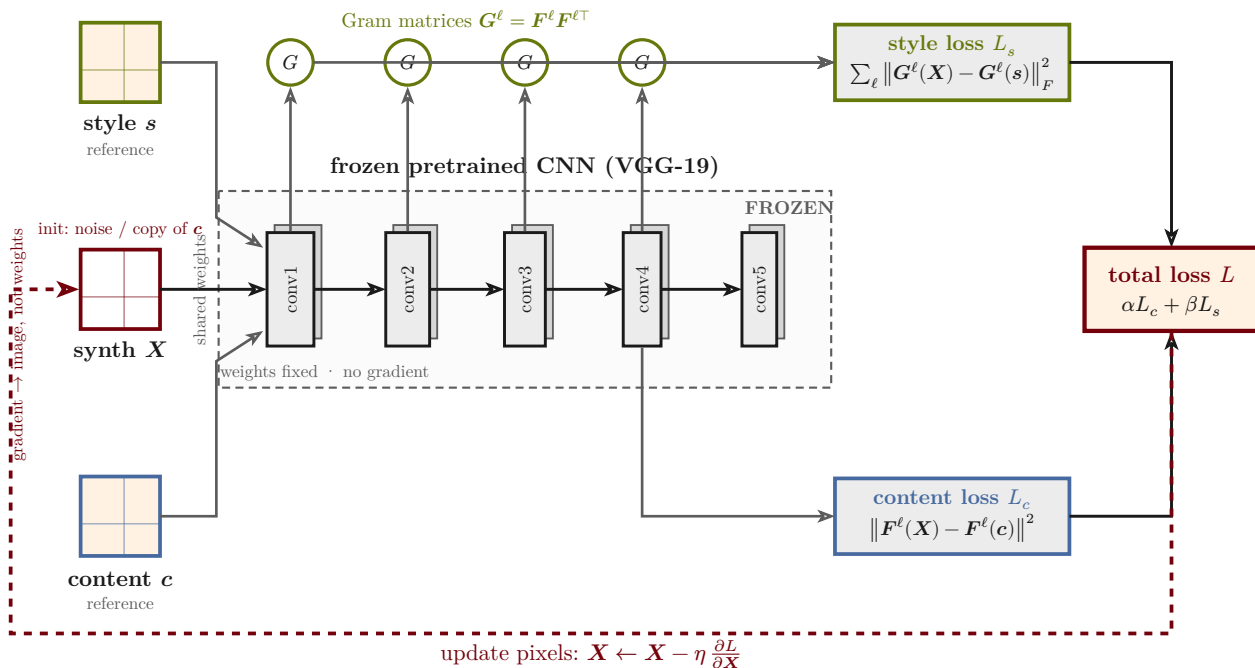


Neural style transfer: optimizing an image through a frozen CNN

one pretrained, **frozen** CNN as a fixed extractor — only the synthesized pixels $\mathbf{X}$ are trained



→ forward

→ feature tap

---> gradient → pixels $\mathbf{X}$

The same frozen CNN extracts features from all three images. **Content** is matched on one deep layer; **style** is matched by the **Gram matrices** (channel-wise feature correlations) across several layers. Only the **synthesized image $\mathbf{X}$** is updated by gradient descent — its pixels are the optimization variables — while the network weights stay fixed.